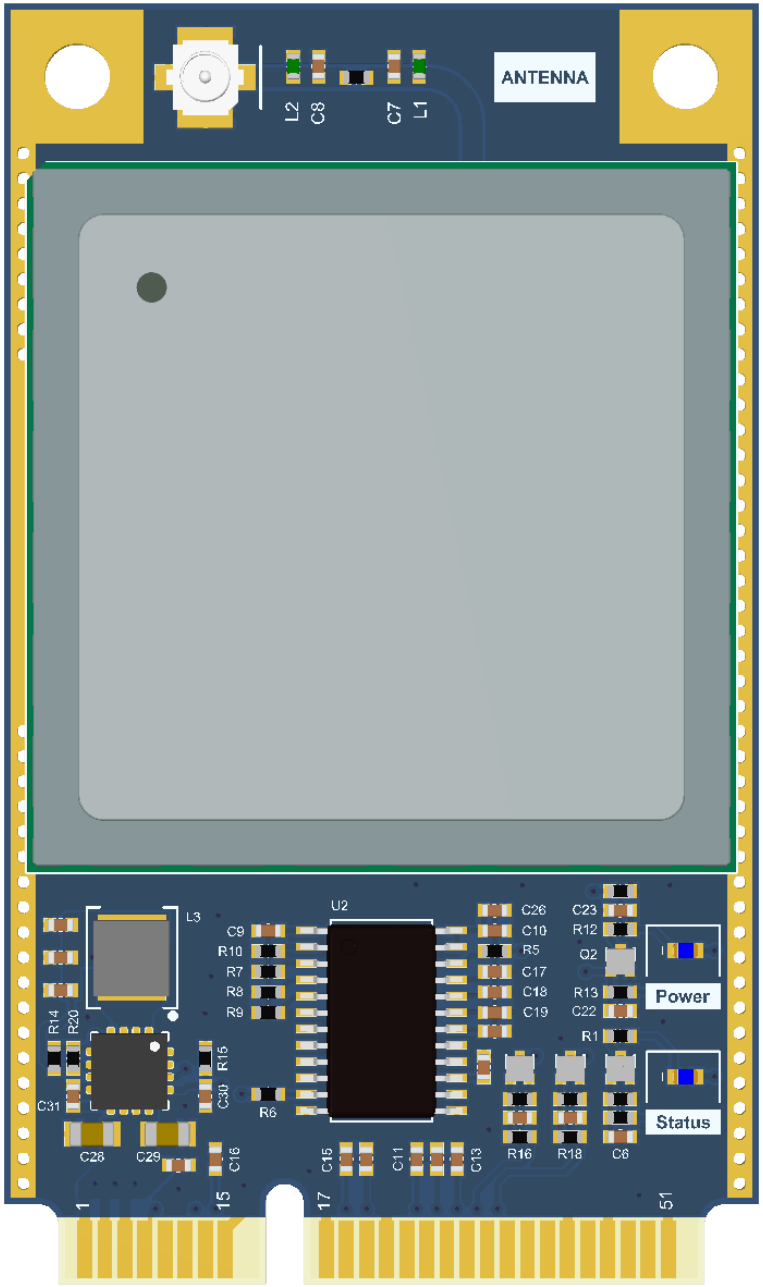
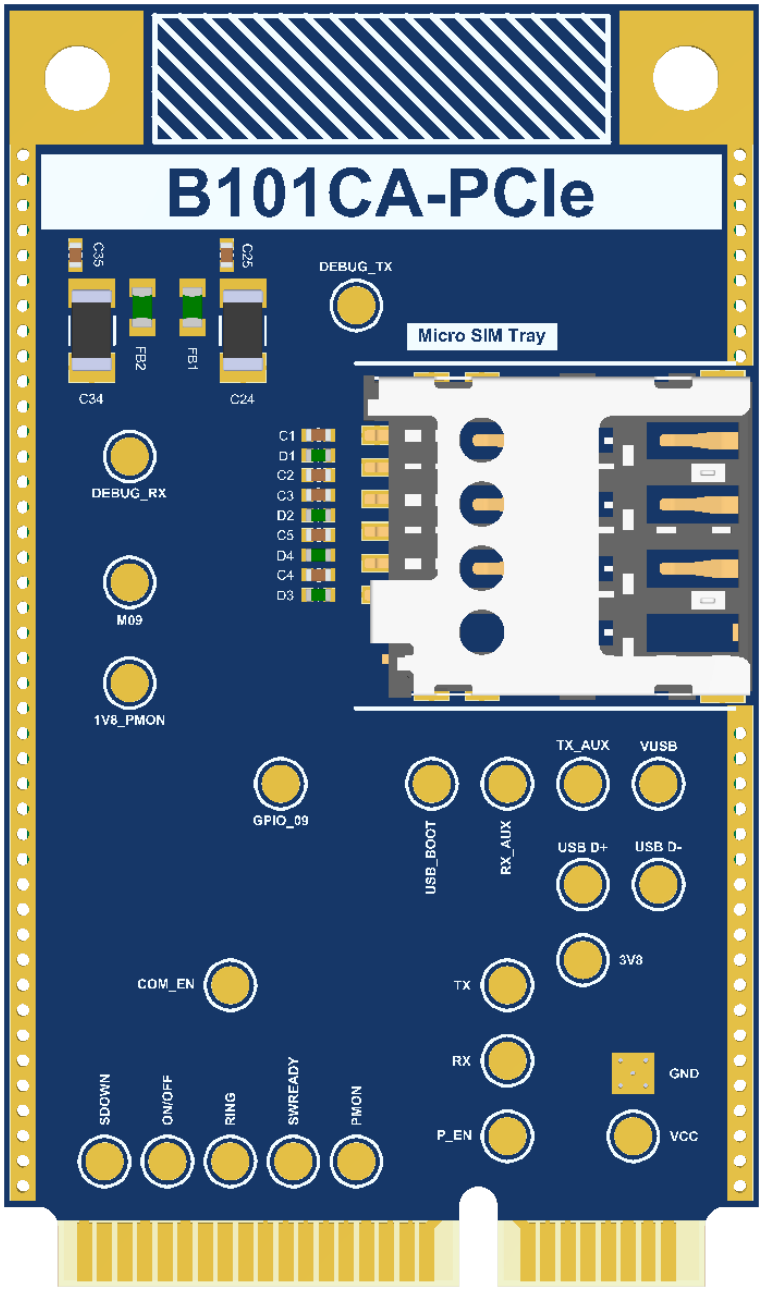
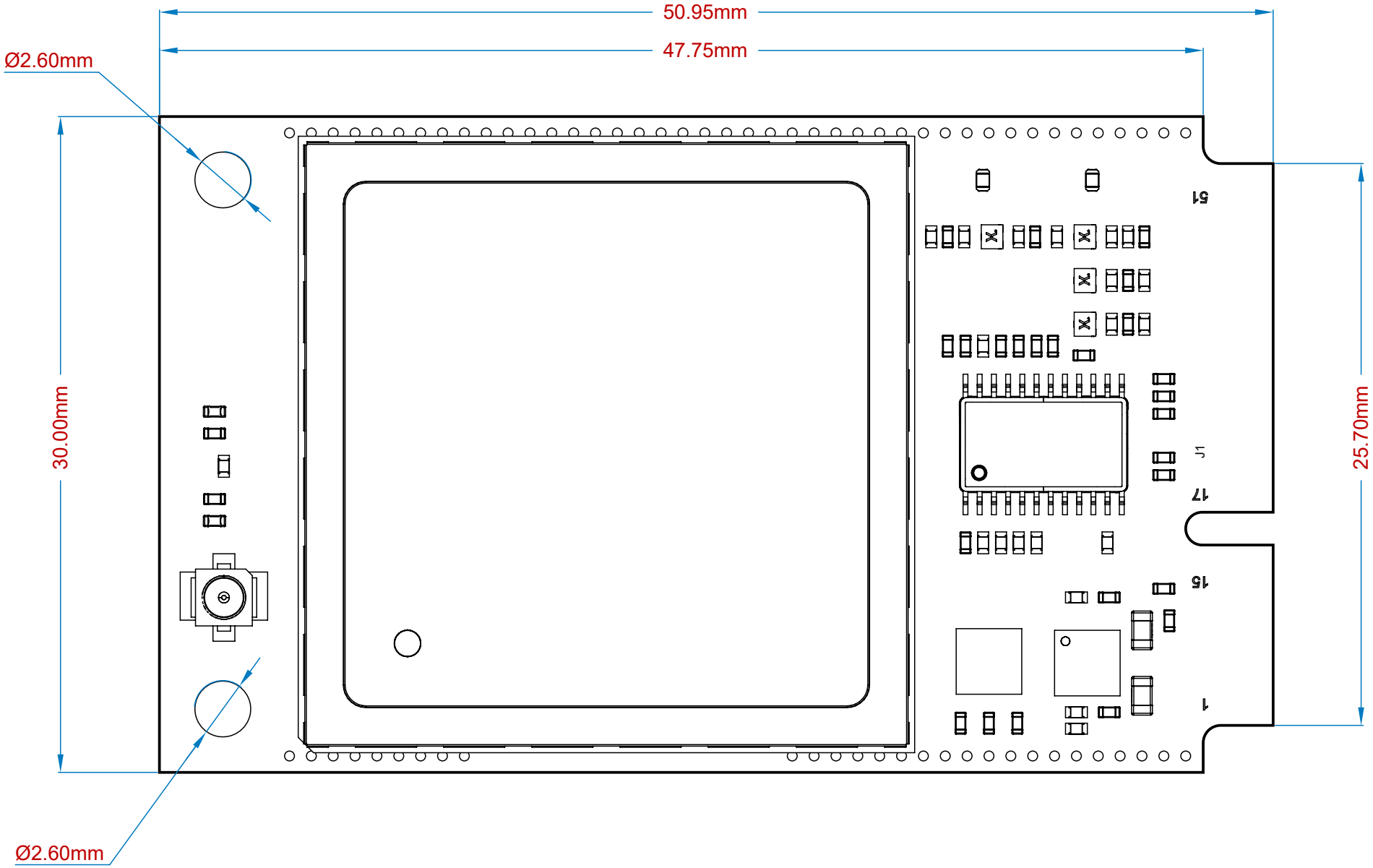
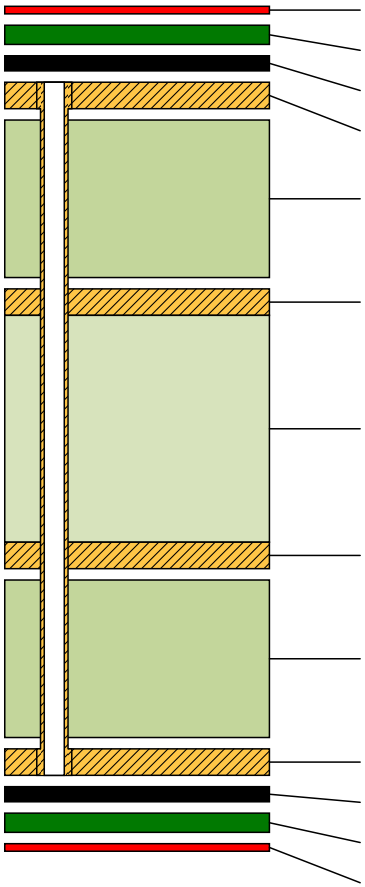

ANTENNA
L2
C8
C7
L1
L3
U2
C9
R10
R7
R8
R9
R14
R20
R15
C30
C31
R6
C28
C29
C16
C15
C11
C13
C26
C10
R5
C17
C18
C19
C23
R12
Q2
R13
C22
R1
R16
R18
C6
Power
Status
1
15
17
51


B101CA-PCIe
C35
C25
DEBUG_TX
Micro SIM Tray
FB2
FB1
C34
C24
C1
D1
C2
C3
D2
C5
D4
C4
D3
DEBUG_RX
M09
1V8_PMON
GPIO_09
USB_BOOT
RX_AUX
TX_AUX
VUSB
USB D+
USB D-
3V3
COM_EN
TX
RX
P_EN
GND
VCC
SDOWN
ON/OFF
RING
SWREADY
PMON


50.95mm
47.75mm
Ø2.60mm
30.00mm
25.70mm
Ø2.60mm
51
J1
17
15
1

Layer Stack Legend

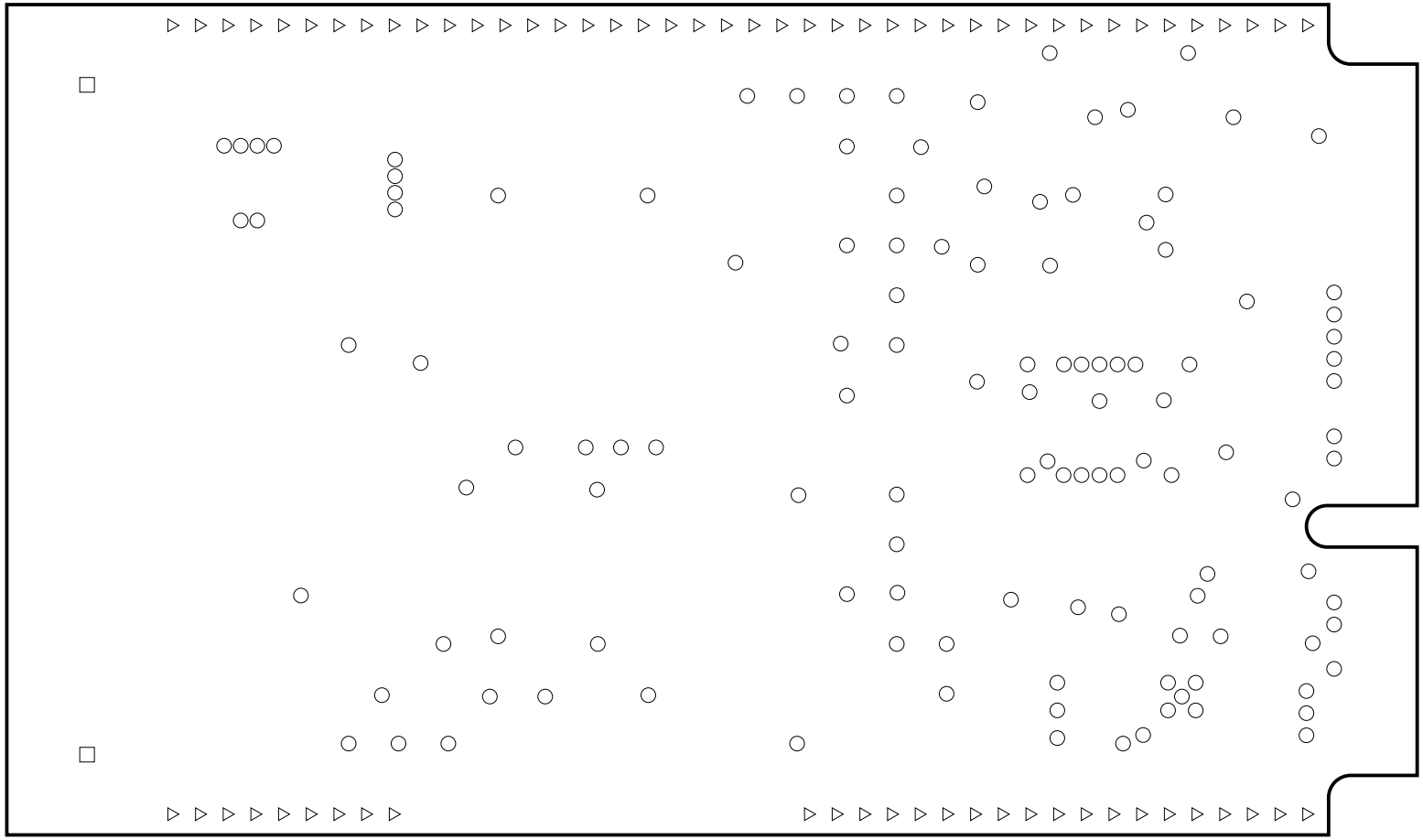


Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.03mm	SM-001	Solder Mask	GTS
Nickel, Gold	Top Copper Plating	0.02mm		Surface Finish	
CF-004	**Top Layer**	**0.04mm**		**Signal**	**GTL**
Prepreg		*0.21mm*	*PP-022*	*Dielectric*	
CF-004	**Int1**	**0.04mm**		**Signal**	**G1**
Core		*0.99mm*	*Core-042*	*Dielectric*	
CF-004	**Int2**	**0.04mm**		**Signal**	**G2**
Prepreg		*0.21mm*	*PP-022*	*Dielectric*	
CF-004	**Bottom Layer**	**0.04mm**		**Signal**	**GBL**
Nickel, Gold	Bottom Copper Plating	0.02mm		Surface Finish	
Surface Material	Bottom Solder	0.03mm	SM-001	Solder Mask	GBS
	Bottom Overlay			Legend	GBO

Total thickness: 1.64mm

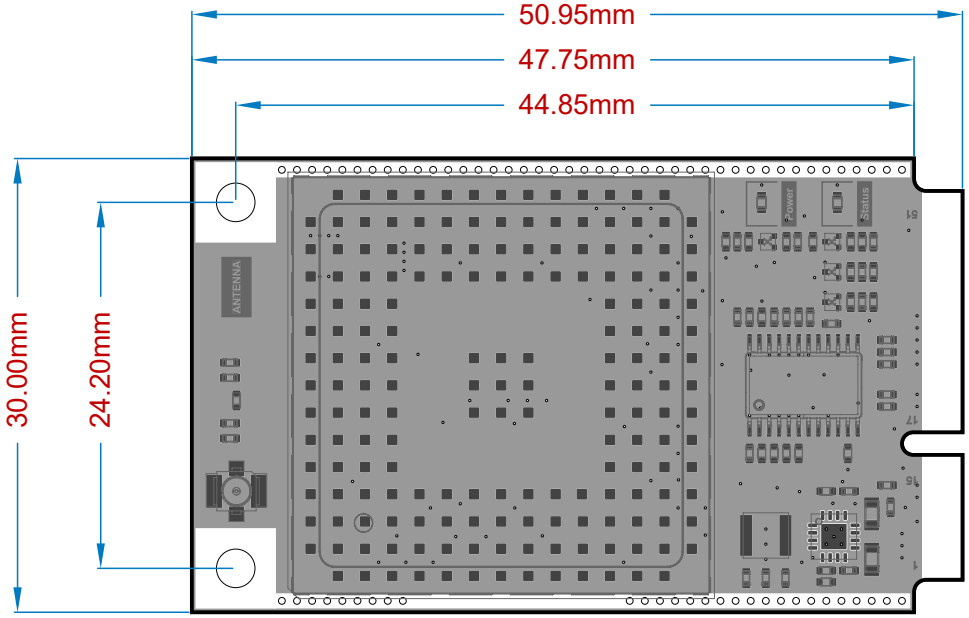
Bill Of Materials

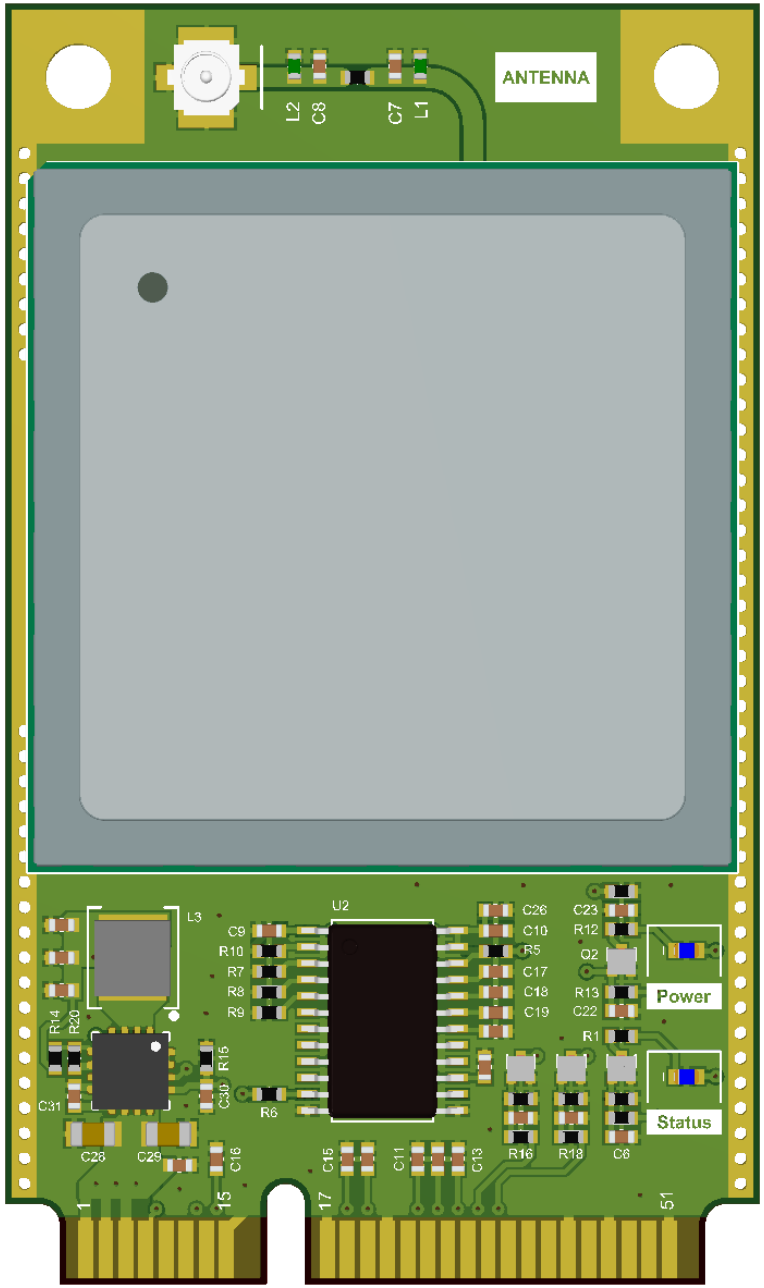
Designator	Quantity	Manufacturer Name	Manufacturer Part Number
C1	1	Murata	GRM153R60J105ME95D
C2, C3, C4, C5, C6, C22, C23, C26	8	Kemet	C0402C330J9RACAUTO
C7, C8	2	AVX	04026C102MAT2A
C9, C10, C33	3	AVX	04026D104KAT4A
C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C25, C30, C35, C36, C37, C38	17	Kemet	C0402C330J9RACAUTO
C24	1	Kemet	C1206C476M8PAC
C27	1	Murata	GCJ188R71E105KA01D
C28, C29	2	Murata Electronics	GRM188R60J226MEA0D
C31	1	KEMET	C0402C472K9RACTU
C32	1	Panasonic	10TPE68M
C34	1	Murata	GRM31CR61A107MEA8L
D1, D2, D3, D4	4	Eaton	0402ESDA-MLP1
D6	1	ROHM Semiconductor	SML-P12MTT86R
D7	1	ROHM Semiconductor	SML-P12VTT86R
FB1, FB2	2	Murata Electronics	BLM18EG101TZ1D
J2	1	GCT	SIM7205-6-0-15-00-A
L1, L2	2	Murata Electronics	LQG15HN1N0S02D
L3	1	Murata	LQH3NPN2R2MMEL
Q1, Q2	2	Nexperia	BC847AQAZ
Q3, Q4	2	Nexperia	BC847AQAZ
R1, R12	2	Vishay	CRCW0402330RFKED
R2	1	Vishay	CRCW04024K70FKED
R3	1	Vishay	CRCW040247K0FKED
R4	1	Vishay	CRCW04020000Z0ED
R5, R6, R7, R8, R9, R10, R15, R17, R19	9	Vishay	CRCW040247K0FKED
R11	1	Vishay	CRCW04020000Z0ED
R13, R20	2	Vishay	CRCW0402100KFKED
R14	1	Vishay	CRCW0402383KFKED
R16, R18	2	Vishay	CRCW040210K0FKED
RF1	1	Molex	73412-0110
U1	1	Telit	LE910C1-EUX
U2	1	Texas Instruments	SN74AXC8T245PWR
U3	1	Texas Instruments	TPS62130ARGTR



Drill Table

Symbol	Count	Hole Size	Plated	Hole Tolerance
○	124	0.20mm	Plated	
▽	70	0.50mm	Plated	+/-0.05mm
□	2	2.60mm	Plated	+/-0.05mm
	196 Total			


50.95mm
47.75mm
44.85mm
30.00mm
24.20mm
ANTENNA
Power
Status


ANTENNA
L2
C8
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L3
U2
C9
R10
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R8
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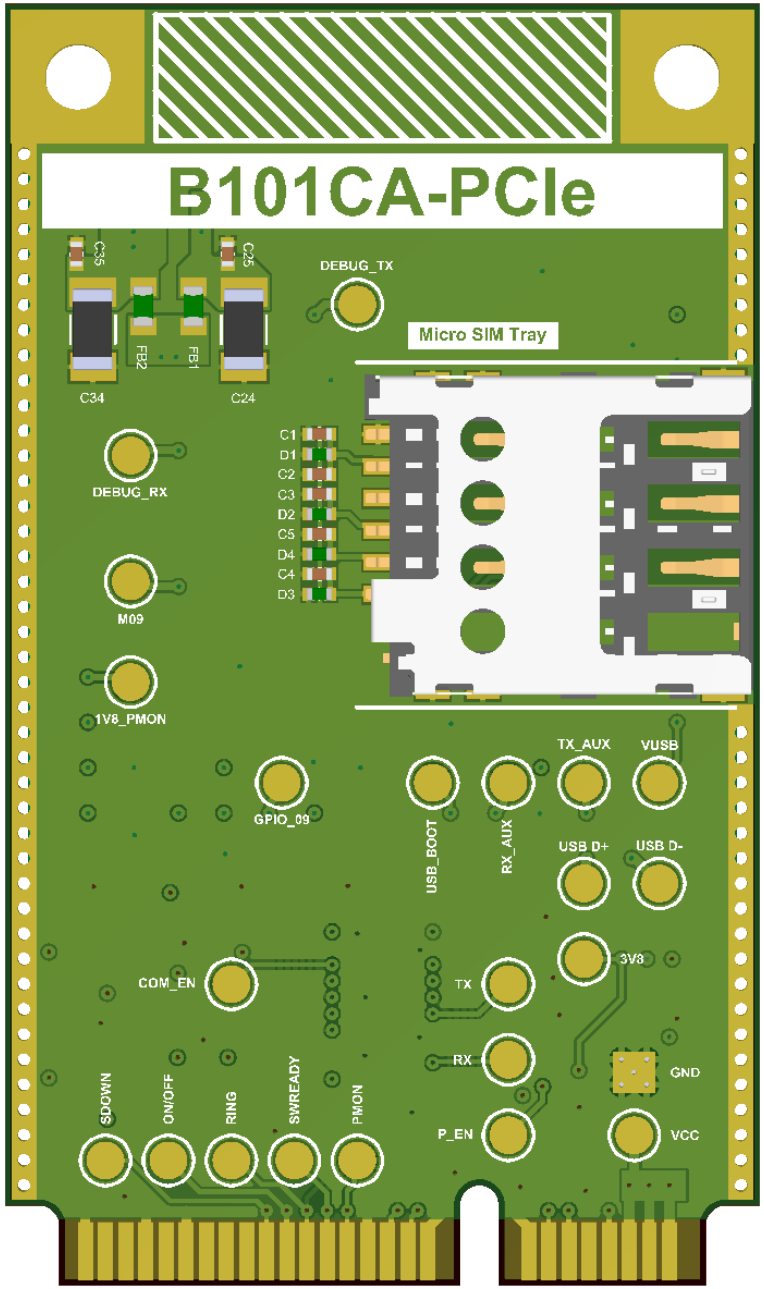
C26
C10
R5
C17
C18
C19
C23
R12
Q2
R13
C22
R1
Power
Status
R14
R20
R15
C30
R6
C31
C28
C29
C16
C15
C11
C13
R16
R18
C6
1
15
17
51


B101CA-PCIe
C35
C25
DEBUG_TX
Micro SIM Tray
FB2
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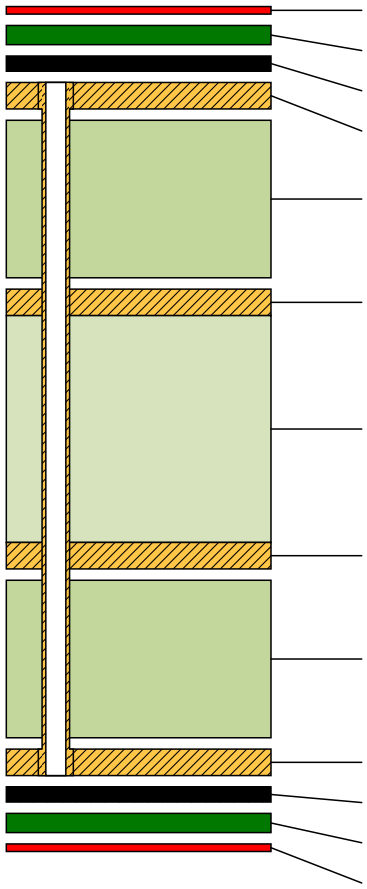
C34
C24
DEBUG_RX
C1
D1
C2
C3
D2
C5
D4
C4
D3
M09
1V8_PMON
GPIO_09
USB_BOOT
RX_AUX
TX_AUX
VUSB
USB D+
USB D-
3V8
COM_EN
TX
RX
P_EN
GND
VCC
SDOWN
ON/OFF
RING
SWREADY
PMON

Layer Stack Legend



Material	Layer	Thickness	Dielectric Material	Type	Gerber
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Prepreg		*0.21mm*	*PP-022*	*Dielectric*	
CF-004	**Int1**	**0.04mm**		**Signal**	**G1**
Core		*0.99mm*	*Core-042*	*Dielectric*	
CF-004	**Int2**	**0.04mm**		**Signal**	**G2**
Prepreg		*0.21mm*	*PP-022*	*Dielectric*	
CF-004	**Bottom Layer**	**0.04mm**		**Signal**	**GBL**
Nickel, Gold	Bottom Copper Plating	0.02mm		Surface Finish	
Surface Material	Bottom Solder	0.03mm	SM-001	Solder Mask	GBS
	Bottom Overlay			Legend	GBO

Total thickness: 1.64mm